

Opinions?? Data Center Water Consumption (AI, Energy, and Data Storage)

Posted by u/JazzlikeAd8934 • 0 points • 6 comments

hi y'all :-) lately i've had this question and i was wondering, how can we really tell how much of what we're doing in our daily lives contribute to this topic? i want to make sure i've got a well grounded environmental sustainability system out here 🙌

Comments

u/KimGo_Eun • 1 points • 2026-02-10 03:17 UTC

I'm also wondering about why some environmental research projects tend to focus on utilizing and maximizing AI-assisted models when AI data centers have this massive effects to the environment too. I need to be enlightened

u/Turbulent_Laugh_4404 • 1 points • 2025-11-28 11:27 UTC

It's a good question because most people only hear about the "energy side" of data centers, but the **water footprint** is just as big a deal, especially now with AI workloads pushing cooling demand way up. The tricky part is that it's almost impossible to know your personal impact because the water use is tied to where the data you're using is actually being processed, and that varies a lot by region and cooling method.

Some data centers use evaporative cooling, which can consume a surprising amount of water, while others rely more on mechanical or liquid cooling and barely use any. This quick breakdown helps explain the different cooling approaches and why the water consumption varies so much:

(<https://eracore.com/data-center-cooling-systems/>)(<https://eracore.com/data-center-cooling-systems/>)

If you're trying to be mindful day to day, the best you can really do is reduce unnecessary cloud usage, storage, or streaming. The rest depends on how efficiently each provider runs their facilities.

u/Daflehrer1 • 1 points • 2025-10-16 20:25 UTC

I watched this video. It documents how data centers affect individuals, communities, and entire regions. I believe it's worthy of careful consideration. It may also fill in gaps in one's knowledge about data centers.

<https://youtu.be/t-8TDOFqkQA?si=wgCqpxfq32UehSYK>

u/Nyobolt • 1 points • 2025-10-01 14:34 UTC

The DRS would lead to reduction of cooling, hence reduction of water consumption.

u/[deleted] • 1 points • 2025-08-25 04:50 UTC

[removed]

u/huffpuffsnuiff • 1 points • 2025-08-23 02:27 UTC

There are new data centre designs that have closed-loop cooling systems where almost no water is lost. The amount is practically negligible.

These cooling systems do trade-off water consumption for power consumption however as they use very power-intensive compressors to cool the water in the system down.

This is obviously not very good unless you are on a green energy source

u/JazzlikeAd8934 • 1 points • 2025-08-25 00:11 UTC

yes, and not every company is spending the money for these systems. its complicated to really tell what these industry companies are doing behind the scenes.

u/huffpuffsnuiff • 2 points • 2025-08-25 00:37 UTC

As an insider I can tell you most aren't. I can see it becoming the standard in country's that are either water scarce or are simply more environmentally conscious. But no, most data centres are being built in America and they do not seem to care too much

u/JazzlikeAd8934 • 1 points • 2025-08-25 00:38 UTC

thank you for your service and intel 🙌

u/p_mindUhealth • 1 points • 2025-08-21 09:34 UTC

Yeah, it's kinda wild data centers use a lot of water for cooling, especially with AI workloads, but companies don't always share the full picture. Location makes a big difference too (a data center in a dry area = bigger impact). On our end, small stuff like lowering streaming quality or choosing more sustainable services helps a bit.

Also, if you're into this topic, there's a global photo essay challenge on sustainability that's open right now could be a fun way to explore it: [\[https://www.collegeaccessprogram.in/sustainability-photoessay-competition/\]](https://www.collegeaccessprogram.in/sustainability-photoessay-competition/)(<https://www.collegeaccessprogram.in/sustainability-photoessay-competition/>)

u/Ksorkrax • 2 points • 2025-08-20 19:41 UTC

A prompt to AI requires about the fraction of a spoon in cooling water. Meanwhile, a sheet of paper requires about twelve liters. In terms of energy, you will use up more energy running your computer than the prompts you can enter in the same time require.

The only issue that should be fixed is that data centers tend to be built horribly. This is however not a necessity, and the same applies to pretty much any field of industry - early complexes built badly until laws are made to prevent that.

u/kozy138 • 1 points • 2025-08-22 04:07 UTC

I think chat gpt is to to a few billions of queries a day. That's a lot of tea spoons...

u/Ksorkrax • 1 points • 2025-08-22 08:06 UTC

Indeed, a lot of teaspoons.

There are also a lot of teaspoon used by all the people in the city by watching Youtube. And a lot of teaspoons, or better to say rivers, used up by creating paper.

If you total them up, you quickly end up with large numbers that tell you about nothing. Let's say you know that a factory uses a million liters per day. Is that much? Is that a low amount? I could not tell.

Therefore, we compare. We take similar things with comparable units. For instance maybe I say I have two hours of time, in which I do some activity that can be described as luxury. Maybe I watch tv, or I draw a picture, or I have AI generate pictures.

And then we compare these tasks. It is quite clear that I will do one of them, and the question is if one of the tasks is particularly wasteful.

If you do it like that and do some math, you will come to the conclusion that no, AI generation isn't particularly wasteful.

For some reason, people in here are very aware that AI has resource costs, but totally ignore the costs of other things, like say watching Youtube.

If you really say that AI image generation or whatever has too high costs for you, then the logical next step is to live like amish people.

Or you spend your efforts more effectively. If you want to reduce your water fingerprint for some reason, lower your meat consumption, especially beef, and buy as few clothes as possible.

However, unless your products are created somewhere where there is dangerously little water, I'd recommend to look for the CO2 fingerprint instead. \[In which AI isn't particularly wasteful compared to other things either.]\

u/JazzlikeAd8934 • 1 points • 2025-08-25 00:10 UTC

i agree with your entire sentiment, up to the part where you claim AI isn't wasteful.

i wouldn't particularly trust AI Company claims that may be bias due to lack of transparency in major corporations wanting to make money from supply and demand. there is a lot of misinformation online, but i've seen a majority of credible information outlets come out with studies that support my claims, and its not just AI thats a problem, its cloud storage (computer and phone storage, streaming services and etc. i'm aware that a majority won't give up these products and that managing our time spent and being mindful about the impacts is one of the best things that we at home can do until we're able to make political/industrial progress. i'll provide a link that mentions this issue https://gradientcorp.com/trend_articles/impacts-of-large-data-centers/

<https://themarkup.org/hello-world/2023/04/15/the-secret-water-footprint-of-ai-technology>

👉 AI can be extremely wasteful especially when millions of users use it daily for things we can learn from other resources (like web browsers that are environmental friendly) but you made a good point about other resources, i pay attention to that as well and have been raising awareness. everything in our lives uses resources and thats why it is important to educate ourselves on said products. i'll attach a water calculator link that has a list of each product and its water consumption intake.

<https://watercalculator.org/footprint/the-hidden-water-in-everyday-products/>

<https://watercalculator.org/water-footprints-101/>

u/Ksorkrax • 1 points • 2025-08-25 00:21 UTC

Uhm... so your sources say that a conversation with ChatGPT (of this and that amount of prompts) requires 500 milliliters and a sheet of paper 5 liters, which is different to the values I am used to but in the same league. So the sheet of paper is ten times worse. Which is my point. You would most likely not tell people that they can't doodle on paper, right?

Again, the question is if you are willing to forgo any luxury that is comparable or worse than AI usage, because that would pretty much mean you'd live like the amish.

u/milkingz69 • 0 points • 2026-01-27 13:53 UTC

"Um, so right? Again, pretty much" This typical redditor needs more gold please!

u/Ksorkrax • 1 points • 2026-01-27 21:58 UTC

You do necromancy on some five month old stuff and then you can't write proper sentences?
Go and get some fresh air mate.

u/Darsich • 1 points • 2026-03-26 19:30 UTC

I mean you are the one who sounds like all of your comments are fed through AI.

u/JazzlikeAd8934 • 1 points • 2025-08-25 00:24 UTC

when you consider how often AI is used by millions worldwide compared to paper (obviously still worldwide but lets say someone buys a journal that lasts them 3 months. then compare that to the millions of AI queries that come in a day.

but you do make a very good point, and its not just AI that contributes to environmental and community decline. its consumerism as a whole and the corporations' methods behind the scenes of mass-producing products

u/OGRITHIK • 1 points • 2026-02-12 08:52 UTC

Apparently, we use [65 billion]
(<https://www.jsinc.com/2022/03/02/facts-about-the-paper-industry/#:~:text=We%20utilize%2065%2C395%2C000%2C000>) sheets of paper daily compared to ChatGPT's 2.5 billion daily queries.

(And yes, GPUs require manufacturing, but a single cluster serves millions of people for years, so that initial environmental cost gets spread out. Paper requires cutting down trees and intense chemical processing for a product that is often used once then tossed.)

When you take that into account, the paper industry is still far worse.

u/JazzlikeAd8934 • 2 points • 2025-08-20 20:00 UTC

just out of curiosity, where did you read that AI takes a tablespoon? i've seen numerous articles and statistics claim otherwise. however either way, you made a lot of very good points. i respect where ur head is at. i forget to talk about consumerism as frequently as i want to sometimes, but thats been on my radar too. commerical flight and shipping definitely. i haven't looked into petrochemicals yet - i'll definitely give that a read.

u/shoulderknees • 1 points • 2025-08-21 07:28 UTC

This is more about 10 tablespoons for a whole page. Mistral release a report about that: <https://mistral.ai/news/our-contribution-to-a-global-environmental-standard-for-ai>

A few things to keep in mind though:

* As mentioned in the study, the model size has a direct impact on the consumption.

* If you look at the current trend in the field, the electrical efficiency (and thus water consumption) dramatically reduces from generation to generation. This means that an evaluation today will be obsolete in a few months. There is at the moment a rebound effect that makes the overall industry impact growth that should be looked at as well though.

u/JazzlikeAd8934 • 1 points • 2025-08-25 00:13 UTC

i understand where ur coming from, however i wouldn't trust a AI specific company to be 100% transparent with the ongoing spread of misinformation online. i've seen numerous studies come out saying otherwise, i'll provide a link to one of them. <https://themarkup.org/hello-world/2023/04/15/the-secret-water-footprint-of-ai-technology>

u/Ksorkrax • 1 points • 2025-08-20 20:35 UTC

Seen it several times but don't remember the exact sites.
I think I remember even the more pessimistic estimates go to only a little bit over a tea spoon.
In any case, nothing close to what paper requires.

